

6(a)	curve drawn to the right of original, same shape and lower maximum rate of reaction, passing through plot of 0.055 to 0.06 mmol dm^{-3} on x-axis ; <i>must start branching away before third plot point at 2au</i> R S-shaped / sigmoid curve $K_m = 0.055$ to $0.06 \text{ mmol dm}^{-3}$; <i>allow ecf if no curve drawn and printed curve used</i> <i>incorrect K_m but standard curve for non-competitive inhibition drawn, allow 2 marks if their K_m is correct if competitive curve drawn <u>and</u> K_m is correct from their curve allow ecf 1 mark</i>	2
6(b)	<i>start codon</i> triplet of / three, (RNA / DNA) bases / nucleotides (begins translation), qualified e.g. coding for, methionine / first amino acid first codon attaching to ribosome or (codon to begin translation is) AUG (for RNA) or TAC (for DNA transcribed strand) or ATG (for DNA non-transcribed strand) ; <i>stop codon</i> triplet of / three, (RNA / DNA) bases / nucleotides, (ending translation), qualified ; <i>codon meaning only needed once</i> e.g. does not code for amino acid stops, chain elongation / addition of amino acids allows / AW, polypeptide to detach (from ribosome) recognised by releasing factors or (codon to end translation) UAG UAA / UGA (for RNA) or ATC / ATT / ACT (for DNA transcribed strand) or TAG / TAA / TGA for DNA non-transcribed strand) <i>if more than one codon given, all must be correct</i>	2

Question	Answer	Marks
6(c)(i)	(each) detection antibody attached to a (different) toxin ; <i>allow one antibody attached to two toxins (astride)</i> A at least one detection antibody attached R if one detection antibody attached directly to immobilised antibody R if unattached detection antibody shown	1
6(c)(ii)	<i>any one from:</i> bind, to different, locations / sites / AW, on toxin ; AW A bind different, epitopes / antigenic determinants detection antibody has, HRP / enzyme (attached) ; ora <i>ignore similarity of both bind toxin to award if stated detection antibody has enzyme</i> capture antibody is immobilised ; ora	1

Question	Answer	Marks
6(c)(iii)	<i>allow enzyme for HRP</i> <i>allow 'lighter' for 'paler'</i> <i>assume ref. to colour is blue-green</i> <i>any four from;</i> 1 colorimeter measures, absorption of light (by reaction mixture) ; A transmission of light (through reaction mixture) 2 low(er) intensity of / pale(r), colour gives low(er), (colorimeter) reading / measurement / absorbance ; ora for transmission A or idea that different intensities of colour, gives different absorbances <i>allow example</i> 3 low(er), intensity of colour / absorbances, indicates low concentration ; A idea that different, intensities of colour / absorbances, indicate different concentrations <i>allow example</i> 4 will get different absorbance readings for low toxin positive result compared with negative result ; AW A would get same reading as blank if negative result (positive does not give same reading as blank) A not relying on by-eye judgement (between colourless and very pale blue-green) A colorimeter not subjective A by-eye judgment may not detect a slight colour change (colourless to very pale blue-green) no ora for statement colorimeter detects a slight colour change (must compare with by-eye) 5,6 AVP ;; e.g. <i>ref. to</i> use of, standards / calibration curve, to obtain actual concentrations <i>ref. to</i> use of, stop / acid, so only initial rate of reaction occurs low concentration of HRP so low quantity of, TMB / substrate, oxidised A less products formed ora low quantity of, oxidised TMB gives a, light(er) (blue-green) colour / minimal colour change ora low concentration of toxin results in, a low quantity of detection antibody binding / most detection antibody washed away	4